Intro to Machine Learning
Practice Problems 3: Logistic Regression and LDA
Practice problems, not turned in — Solutions

Directions: These practice problems go with Week 3 (logistic regression, including the multinomial model, and linear discriminant analysis). Problems 7 and 8 review ridge regression and the lasso from Week 2. They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Problems adapted from ISL cite the exercise; see the book for the original wording. Practice Problems 4 covers QDA and naive Bayes.

Notation follows the lecture. Classes are $k = 1, \dots, K$, $\pi_k = \Pr(Y = k)$ is the prior, $f_k(x)$ is the density of X in class k, and $p_k(x) = \Pr(Y = k \mid X = x)$ is the posterior. When fitting LDA by hand, use the pooled variance with divisor $n - K$.

1. **A fitted logistic model.** (ISL 4.8, Exercise 6; ISL 4.3.) Data on students in a statistics class give $x_1 =$ hours studied, $x_2 =$ undergraduate GPA and $Y = 1$ if the student gets an A. A logistic regression gives $\hat{\beta}_0 = -6$, $\hat{\beta}_1 = 0.05$ and $\hat{\beta}_2 = 1$.

   (a) Estimate the probability that a student who studies for 40 hours and has a GPA of 3.5 gets an A.

   (b) How many hours would that student need to study to have a 50% chance of an A?

   (c) With a 0.5 threshold, the students predicted to get an A are on one side of a line in the (hours, GPA) plane. Write the line as hours $= a + b\times$ GPA. How many hours does a student with a 3.0 GPA need?

   Solution.

   (a) The linear predictor is $\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 = -6 + 0.05(40) + 1(3.5) = -0.5$, so

   $$\hat{p} = \frac{e^{-0.5}}{1 + e^{-0.5}} = \frac{0.607}{1.607} = 0.378.$$

   (b) $\hat{p} = 0.5$ exactly when the linear predictor is 0 (then $\hat{p} = e^0/(1+e^0)$): $-6+0.05h+3.5 = 0$, so $h = 2.5/0.05 = 50$ hours.

   (c) The boundary is $-6 + 0.05\,\text{hours} + \text{GPA} = 0$, that is hours $= 120 - 20 \times$ GPA. Students above the line (more hours for their GPA) are predicted to get an A. With a 3.0 GPA, $120 - 60 = 60$ hours.

2. **Multinomial logistic regression.** (ISL 4.3.5.)

(a) Commuters choose to walk, bike or drive, and x is the distance to work in km. A multinomial logistic regression with *drive* as the baseline gives

$$\hat{p}_{\text{walk}}(x) = \frac{e^{4-2x}}{D(x)}, \qquad \hat{p}_{\text{bike}}(x) = \frac{e^{2-0.5x}}{D(x)}, \qquad \hat{p}_{\text{drive}}(x) = \frac{1}{D(x)},$$

where $D(x) = 1 + e^{4-2x} + e^{2-0.5x}$. Find the three fitted probabilities for a commuter who lives 2 km away.

(b) At what distance are biking and driving equally likely? Walking and biking? Which mode is predicted for very large x?

(c) How many coefficients does this model have? How many would it have with $p = 4$ predictors and $K = 5$ modes?

Solution.

(a) At $x = 2$, $e^{4-4} = 1$ and $e^{2-1} = e$, so $D(2) = 2 + e = 4.718$ and

$$\hat{p}_{\text{walk}} = \frac{1}{2+e} = 0.212, \qquad \hat{p}_{\text{bike}} = \frac{e}{2+e} = 0.576, \qquad \hat{p}_{\text{drive}} = \frac{1}{2+e} = 0.212.$$

(b) All three share the denominator $D(x)$, so compare numerators. Bike and drive tie when $e^{2-0.5x} = 1$, that is $2 - 0.5x = 0$, at $x = 4$ km. Walk and bike tie when $e^{4-2x} = e^{2-0.5x}$, that is $4 - 2x = 2 - 0.5x$, at $x = 4/3$ km. For large x both exponents go to $-\infty$, so the walk and bike numerators go to 0 and driving is predicted. (Every pairwise boundary is a single point here; with more predictors they are hyperplanes.)

(c) $(K-1)(p+1) = 2 \cdot 2 = 4$. With $p = 4$ and $K = 5$: $4 \cdot 5 = 20$.

3. **The LDA discriminant and its boundary.** (ISL 4.8, Exercise 2; ISL 4.4.1–4.4.2, ESL 4.3.)

(a) Take $p = 1$ and $X \mid Y = k \sim N(\mu_k, \sigma^2)$, with the same σ^2 in every class. Show that the Bayes classifier assigns x to the class with the largest

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k.$$

(b) Now take $p \geq 1$, $X \mid Y = k \sim N(\mu_k, \Sigma)$, and the lecture's $\delta_k(x) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log \pi_k$. Show that the boundary between classes k and l, $\{x : \delta_k(x) = \delta_l(x)\}$, is a hyperplane $\{x : a + x^T b = 0\}$, and give a and b.

(c) With $K = 3$ and $p = 4$, how many parameters does LDA estimate (priors, means and the shared covariance)? How many coefficients does multinomial logistic regression estimate? The LDA posterior has the same form as a multinomial logistic regression (lecture §2.2), so what is LDA doing with the extra parameters?

Solution.

(a) The Bayes classifier picks the k with the largest posterior, $p_k(x) = \pi_k f_k(x) / \sum_l \pi_l f_l(x)$. The denominator doesn't depend on k, so it picks the largest $\pi_k f_k(x)$, or equivalently the largest log:

$$\log \pi_k + \log f_k(x) = \log \pi_k - \log\left(\sqrt{2\pi}\,\sigma\right) - \frac{(x-\mu_k)^2}{2\sigma^2}$$

$$= \log \pi_k - \log\left(\sqrt{2\pi}\,\sigma\right) - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}.$$

The terms $-\log(\sqrt{2\pi}\,\sigma)$ and $-x^2/(2\sigma^2)$ are the same for every class (because σ is shared), so they don't change which k is largest. Dropping them leaves $\delta_k(x)$.

(b) $\delta_k(x) - \delta_l(x) = x^T\Sigma^{-1}(\mu_k - \mu_l) - \frac{1}{2}\left(\mu_k^T\Sigma^{-1}\mu_k - \mu_l^T\Sigma^{-1}\mu_l\right) + \log(\pi_k/\pi_l)$, and the boundary is where this is 0. So

$$b = \Sigma^{-1}(\mu_k - \mu_l), \qquad a = \log\frac{\pi_k}{\pi_l} - \tfrac{1}{2}(\mu_k + \mu_l)^T\Sigma^{-1}(\mu_k - \mu_l).$$

(The two forms of the constant are equal because Σ^{-1} is symmetric, so the cross terms $\mu_k^T\Sigma^{-1}\mu_l$ and $\mu_l^T\Sigma^{-1}\mu_k$ cancel.) The equation $a + x^Tb = 0$ is linear in x, so the boundary is a hyperplane: a line when $p = 2$.

(c) LDA: $K - 1 = 2$ priors, $Kp = 12$ means and $p(p+1)/2 = 10$ covariance entries, so 24 parameters. Logistic regression: $(K-1)(p+1) = 10$. Both end with a posterior that has 10 free coefficients (the same form, lecture §2.2), but LDA also models how X is distributed, which logistic regression leaves alone. If that model is right, the extra structure lets LDA use every observation to estimate the coefficients, so it has lower variance. If it is wrong, the coefficients are biased (lecture §5; the LDA vs. logistic lab, Sections 2 and 3).

4. **LDA with one predictor.** (ISL 4.8, Exercise 7; ISL 4.4.2.) We want to predict whether a stock will issue a dividend this year (yes or no) from X, last year's percentage profit. Among companies that issued a dividend, X averaged $\bar{x} = 10$; among those that didn't, $\bar{x} = 0$. The variance of X in both groups is $\hat{\sigma}^2 = 36$, and 80% of companies issued dividends. Assume X is normal in each group. Find the probability that a company with $X = 4$ issues a dividend this year.

Solution. By Bayes' theorem, with the normal density's constant $1/(\sqrt{2\pi}\,\sigma)$ cancelling,

$$\hat{\Pr}(\text{dividend} \mid X = 4) = \frac{0.8\, e^{-(4-10)^2/72}}{0.8\, e^{-(4-10)^2/72} + 0.2\, e^{-(4-0)^2/72}}$$

$$= \frac{0.8\, e^{-0.5}}{0.8\, e^{-0.5} + 0.2\, e^{-0.222}} = \frac{0.4852}{0.4852 + 0.1601} = 0.752.$$

Even though 4 is closer to the "no" mean of 0 than to 10, the 4-to-1 prior for "yes" wins.

5. **Fitting LDA by hand.** (ISL 4.4.2.) A training set has one predictor:

$$\text{class 1: } x = 1, 2, 3; \qquad \text{class 2: } x = 5, 6, 6, 7.$$

(a) Compute $\hat{\pi}_k$, $\hat{\mu}_k$ and the pooled variance $\hat{\sigma}^2$.

(b) Find the LDA decision boundary. Which way did the unequal priors move it from the midpoint of the means?

Solution.

(a) $n = 7$, $K = 2$. $\hat{\pi}_1 = 3/7$, $\hat{\pi}_2 = 4/7$; $\hat{\mu}_1 = 2$, $\hat{\mu}_2 = 6$. The within-class sums of squares are $1 + 0 + 1 = 2$ and $1 + 0 + 0 + 1 = 2$, so $\hat{\sigma}^2 = (2 + 2)/(7 - 2) = 0.8$.

(b) From Problem 3(a), $\delta_1(x) = 2x/0.8 - 4/1.6 + \log(3/7)$ and $\delta_2(x) = 6x/0.8 - 36/1.6 + \log(4/7)$. Setting them equal, $5x = 20 + \log(3/4)$, so $x^* = 4 - 0.058 = 3.942$. Class 2 has the larger prior, so the boundary moves toward class 1, the rarer class, and class 2 claims a little more of the line.

6. **LDA or logistic regression?** (ISL 4.5; lecture §5; the LDA vs. logistic lab.) For each situation, say which you would try first and why.

(a) $n = 40$, $p = 3$, and histograms of each predictor look roughly normal, with about the same spread in both classes.

(b) $n = 50{,}000$, and most predictors are binary indicators or heavily skewed counts.

(c) The two classes are perfectly separated in the training data, and you need probabilities.

(d) One class contains a handful of extreme values far from the other class.

(e) True or false: since LDA and logistic regression have posteriors of the same form, fitting both to the same data gives the same coefficients.

Solution.

(a) **LDA.** Its assumptions look about right, and with small n its lower variance matters. When the Gaussian model is true, logistic regression needs noticeably more data to reach the same accuracy (in the lab, LDA's excess error was about half of logistic regression's).

(b) **Logistic regression.** LDA's Gaussian model is clearly wrong for binary and skewed predictors, and that bias doesn't shrink as n grows. With $n = 50{,}000$, LDA's variance advantage is worth little.

(c) **LDA**, or a penalized logistic regression. With separated classes the unpenalized logistic MLE doesn't exist (DSAI 601), and its probabilities go to 0 and 1. LDA's estimates exist whatever the separation.

(d) **Logistic regression.** Points far from the boundary barely affect its fit, but they move LDA's means and inflate its covariance.

(e) **False.** The two use the same *form* of model but estimate it differently: LDA from $\hat{\pi}_k$, $\hat{\mu}_k$ and $\hat{\Sigma}$, logistic regression by maximizing the conditional likelihood. On a finite sample the coefficients differ, though they are often close when the Gaussian model fits.

7. **Ridge and lasso shrinkage.** (ISL 6.2.2.) In the special case of ISL 6.2.2 ($n = p$, $\mathbf{X}$ the identity matrix, no intercept), least squares gives $\hat{\beta}_j = y_j$, and ridge and the lasso have closed forms:

$$\hat{\beta}_j^R = \frac{y_j}{1 + \lambda}, \qquad \hat{\beta}_j^L = \begin{cases} y_j - \lambda/2 & \text{if } y_j > \lambda/2, \\ y_j + \lambda/2 & \text{if } y_j < -\lambda/2, \\ 0 & \text{if } |y_j| \le \lambda/2. \end{cases}$$

(a) With $y = (3, -1.5, 0.8, -0.2)$ and $\lambda = 2$, compute the ridge and lasso estimates.

(b) Which method set coefficients exactly to zero? Which shrank the largest coefficient the most?

(c) What do both estimates become as $\lambda \to 0$? As $\lambda \to \infty$?

Solution.

(a) Ridge divides each y_j by $1 + \lambda = 3$: $(1, -0.5, 0.267, -0.067)$. The lasso moves each y_j toward zero by $\lambda/2 = 1$, and sets it to zero if $|y_j| \leq 1$: $(2, -0.5, 0, 0)$.

(b) Only the lasso set coefficients to zero (the third and fourth). Ridge shrank the largest coefficient more: from 3 to 1, a drop of 2, while the lasso took off only 1. Ridge shrinks by the same *proportion*, so big coefficients lose the most; the lasso shrinks by the same *amount* and zeroes out the small ones.

(c) As $\lambda \to 0$ both become y_j, the least squares estimates. As $\lambda \to \infty$ both go to 0. Ridge gets there only in the limit, while the lasso is exactly 0 for every coefficient once $\lambda \geq 2 \max_j |y_j|$ (here $\lambda \geq 6$).

8. **Ridge or the lasso?** (ISL 6.2.3.) For each situation, say which you would try first and why.

(a) There are 200 predictors, and you believe only a handful of them affect the response.

(b) There are 200 predictors, and you believe most of them have small effects.

(c) You need a model with only a few predictors that you can explain to a manager.

(d) True or false: before fitting ridge or the lasso, you should standardize the predictors.

Solution.

(a) **The lasso.** It can set the unimportant coefficients exactly to zero, so it does variable selection and matches a sparse truth. Ridge would keep all 200 in the model.

(b) **Ridge.** If nearly every predictor matters a little, setting coefficients to zero throws away real signal. Ridge keeps them all and shrinks them, which reduces variance without forcing sparsity (ISL 6.2.3 shows ridge winning in this setting).

(c) **The lasso**, because its model contains only the predictors with nonzero coefficients. Ridge's model always contains all of them.

(d) **True.** The penalty treats every β_j the same, but the size of β_j depends on the units of x_j (measuring x_j in grams instead of kilograms divides its coefficient by 1000). Without standardizing, the amount each predictor is shrunk would depend on its units.